

Network Access: Home Network Audit

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Discipline: Offensive and Defensive Cybersecurity

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1. Test Information

This document details the implementation of Laboratory 3 - Network Access, inserted in the Cybersecurity course unit Offensive and Defensive. The central objective of this laboratory is to provide practical examples of the use of network access and mapping with a view to developing a set of safety tests.

Network access control is the first line of defense for any technological infrastructure, this practical exercise allows Consolidate knowledge about the tactics used by attackers in the early stages of an intrusion, as well as defensive measures necessary to mitigate them.

1.1. Scope and Environment of Testing

To ensure full legal and ethical compliance of this assessment (operating on a *White-Hat* model), the target selected for this lab was the author's own local network and home infrastructure.

- **Main Target:** Private Home Network (LAN and WLAN).
- **Internal Network Segment:** [IP Redacted]
- **Limitations (Exclusions):** Because it is a home environment and to focus the evaluation on the strictly technical component, the social engineering tests related to network access were not performed. The physical inspection of the network was limited to checking the equipment provided by the ISP (Router/Gateway)

2. Network Recognition and Discovery (Scanning)

The initial phase of network access focuses on mapping the infrastructure to carry out the collection of information on the network and your devices. The main objective is to understand the local topology, discover active systems and identify ports (TCP and UDP).

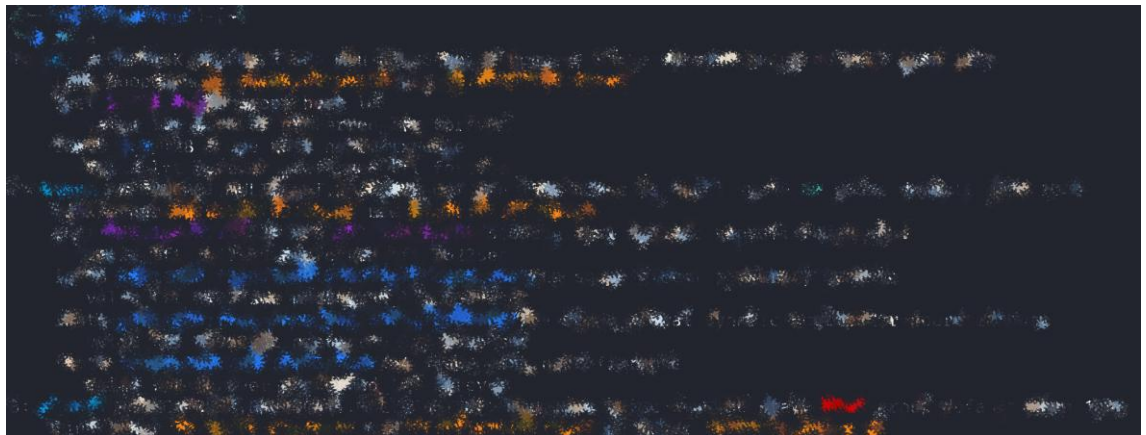
2.1. Identification of Connection and Interfaces

To start the audit with context about the source network, the public IP address was identified and the local network interfaces were surveyed to find out the private IP assigned to the attacking machine.

Command used to check Public IP:



Command used to check Private IP and Interfaces:



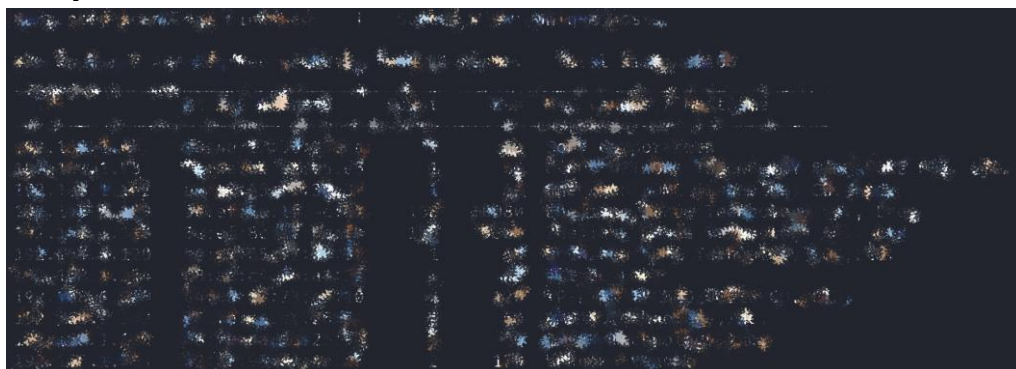
2.2. Device Discovery (Host Discovery)

With the network segment identified, different approaches were used to detect the active systems on the local network. Tools that perform sweeps through ARP and ICMP requests were used.

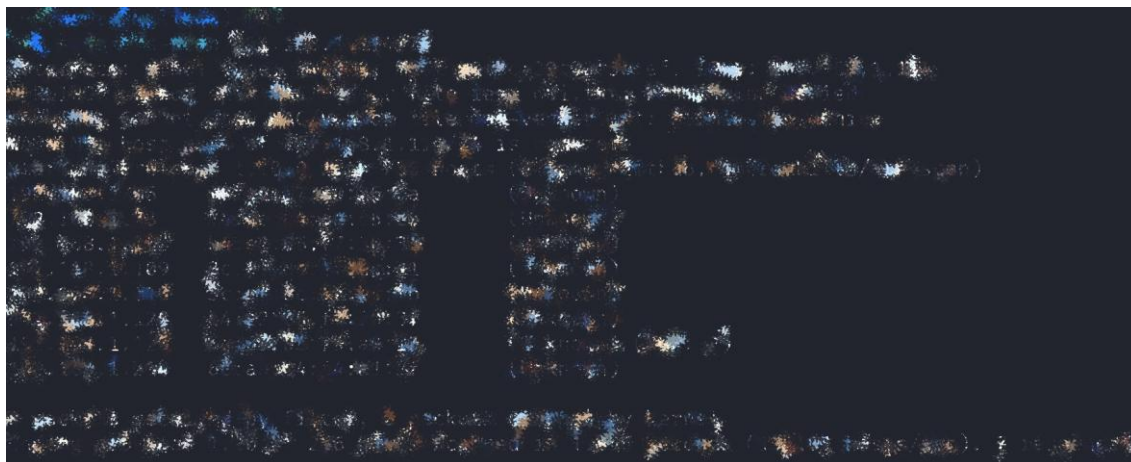
Command used (Netdiscover - ARP Scan):

```
sudo netdiscover -r [IP Redacted]
```

Output:



Command used (ARP-Scan):



Command used (Nmap - Ping Scan):

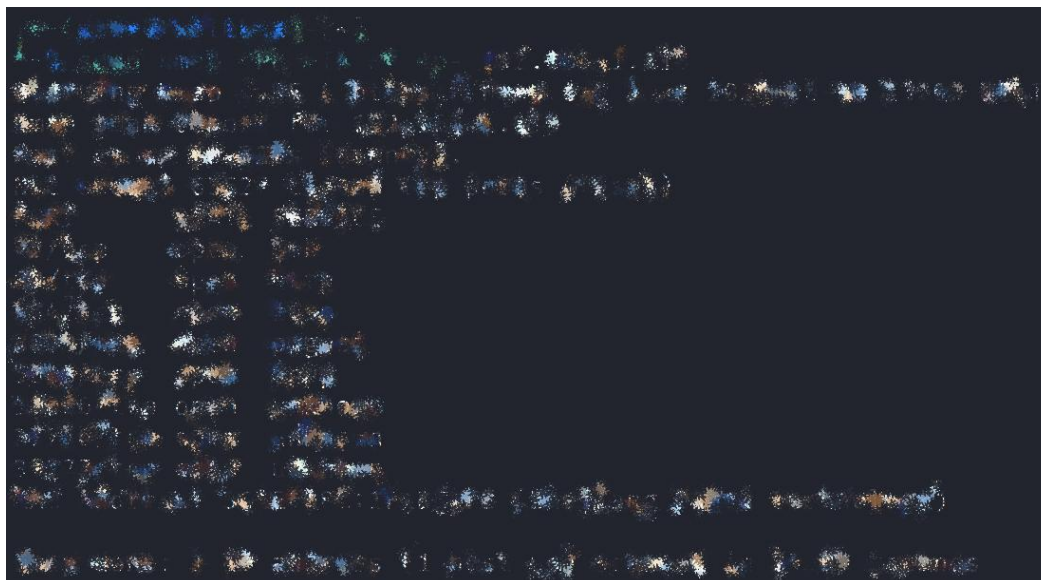


2.3. Port Scanning

After identifying the *active hosts*, a specific target was selected in the network to check the state of the ports (open, closed or filtered) and infer the services running.

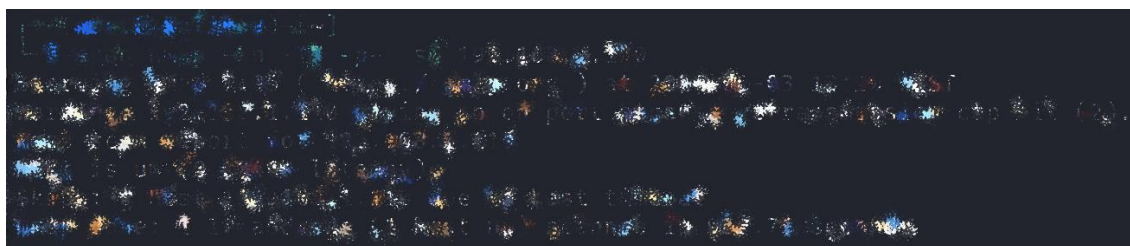
Command used (Nmap TCP Full with packet fragmentation):

Using the `-f` flag allows you to fragment packets to try to bypass detection mechanisms (IDS) or firewalls.



```
[root@kali:~]# nmap -f 192.168.1.10
Nmap scan report for 192.168.1.10
Host is up (0.0000000s latency).
Not shown: 65534 closed ports
PORT      STATE SERVICE
22/tcp    OPEN  SSH
80/tcp    OPEN  HTTP
443/tcp   OPEN  HTTPS
8080/tcp  OPEN  HTTP
10000/tcp OPEN  LISTENING
```

Command used (Full UDP Nmap): Scanning directed to UDP ports (e.g. DNS, DHCP).



```
[root@kali:~]# nmap -sU 192.168.1.10
Nmap scan report for 192.168.1.10
Host is up (0.0000000s latency).
Not shown: 65534 closed ports
PORT      STATE SERVICE
53/udp    OPEN  DNS
67/udp    OPEN  DHCP
10000/tcp OPEN  LISTENING
```

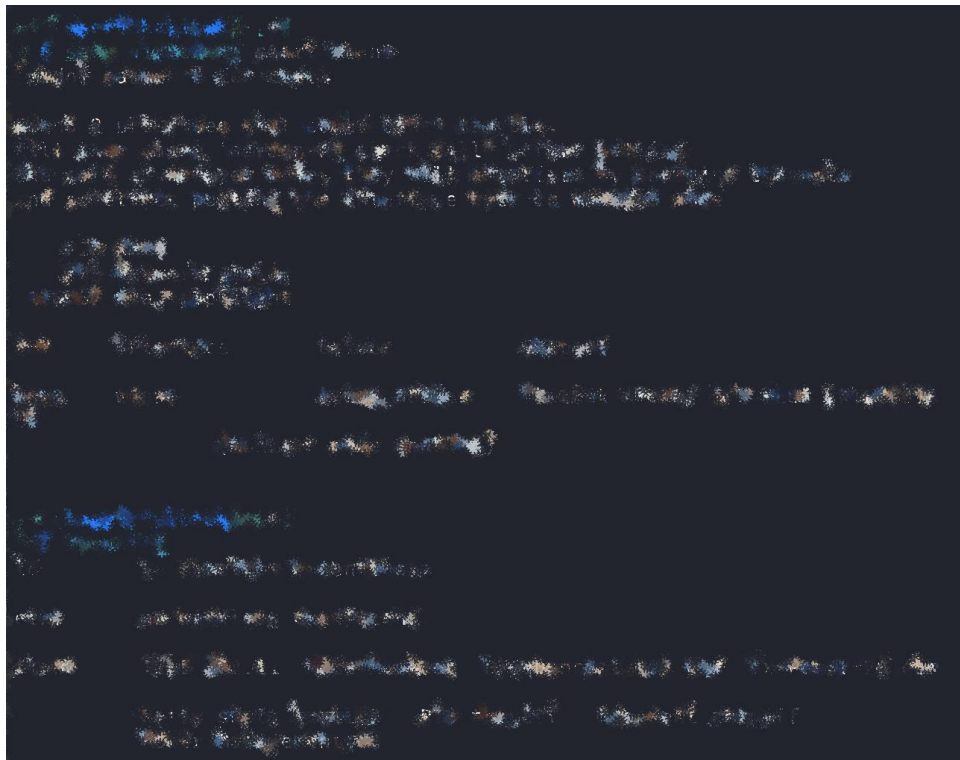
3. Wireless Access Tests

At this stage of the audit, the focus is on the radio spectrum, in order to identify available wireless networks, their authentication mechanisms and testing potential vectors of Infrastructure entry.

3.1. Preparation and Monitoring (Monitor Mode)

To intercept the surrounding 802.11 traffic, it was necessary to identify the available wireless network interfaces and change the operation of the network adapter to monitor mode.

Commands used to configure the interfaces:



3.2. Network Mapping (Airodump-ng)

With the interface in monitor mode, passive capture was carried out to identify the surrounding networks, extracting vital information such as the SSID, BSSID (Access Point MAC Address), channels in use and encryption types.

Command used (Global search in bands a/b/g):



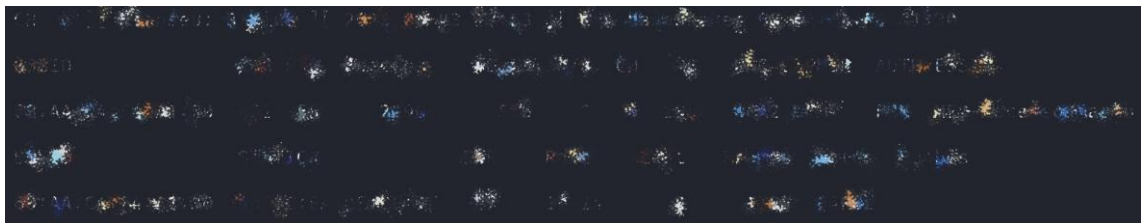
3.3. Targeted Capture

After identifying the target network, the scan was isolated to a specific channel and BSSID. This allows capturing traffic directed only to that equipment and saving the packets in a file (.cap) for later analysis or attempted credential extraction.

Command used (Capture focused on a specific BSSID and Channel):

```
sudo airodump -ng -c 6 --bssid [IP Redacted] wlan0 -  
w capture-wep
```

Output:



3.4. Automated Auditing (Wifite)

As a complementary validation to the security of the WPA/WPA2 network, an automated framework was used . This tool orchestrates various attacks (such as capturing *handshakes*) to test for common vulnerabilities in wireless infrastructure.

Commands used (Wifite targeting WPA networks and a specific channel):



4. Traffic Inspection and Sniffing

Network access auditing is incomplete without network access analysis traffic that circulates in it. The purpose of this section is to capture data in transit (*Sniffing*) to identify the protocols used, extract files or credentials transmitted in clean text and detect anomalies in communication.

4.1. ARP Health Check (MitM Detection)

In a switched network, one of the most common attacks to intercept third-party traffic is ARP *Poisoning/Spoofing* (ARP *Poisoning/Spoofing*) tables. Before starting packet capture, the ARP table of the attacking machine was checked for duplicate MAC Addresses (which would indicate an ongoing *Man-in-the-Middle* attack).

Command used (ARP Check):

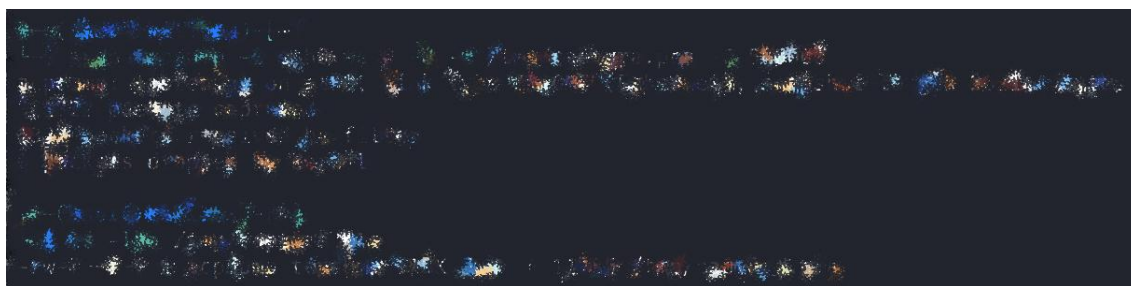


4.2. Network Packet Capture (Tcpdump)

To ensure that the analysis was not limited to real-time data, a command-line tool was used to capture the traffic from the main interface and save it in a .pca (*packet capture*) file. This file serves as forensic evidence.

Commands used (Capture and Verification):

tcpdump was configured to capture full-size packets (-s 0) on the eth0 interface and split them into files if they reached a certain size.



4.3. Graphical Analysis and Artifact Extraction

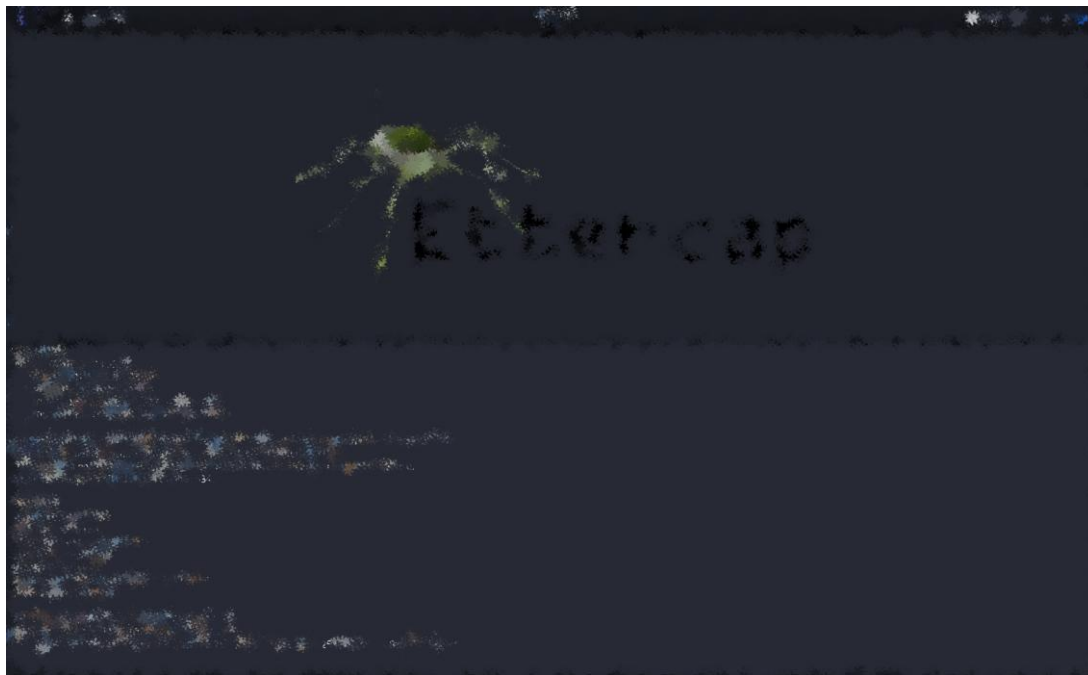
With the traffic captured, graphical tools were used to dissect the packets and extract intelligible information.

Command used (Ettercap):

Ettercap is a tool designed for *Unified Sniffing* and MitM attacks. Its graphical environment has been invoked to allow *host mapping* and active listening.

```
sudo ettercap -G
```

Output:



Command used (Package Analysis with NetworkMiner):

NetworkMiner was used to read the previously captured .pca file. This tool allows you to extract files, credentials, images, and sessions in clean text in an automated way.

```
sudo mono /opt/NetworkMiner_3-0/NetworkMiner.exe
```


Output:



As shown in the image above, passive analysis with NetworkMiner allowed the identification of 53 hosts, fingerprinting operating systems (Linux, Windows, macOS) and intercepting traffic from external applications (such as Microsoft Teams).

5. Conclusions and Recommendations

The realization of this laboratory in the home network infrastructure allowed the application, in a practical and controlled way, of the methodologies of key recognition, interception and auditing in the area of *Network Access*.

Through the execution of the tests, it was possible to extract the following conclusions on the security posture of the network, followed by the respective mitigation recommendations:

5.1. Network and Attack Surface Visibility

Conclusion: Passive and active scanning tools (netdiscover, arp-scan, nmap) proved to be highly effective in mapping the network topology, quickly identifying active *hosts* and exposed ports. Internal visibility is total for any device connected to the local network.

Recommendation: To mitigate the risk of lateral movements in the event of a device being compromised (e.g. a mobile phone or PC), network segmentation is recommended. Creating a guest network (*Guest Wi-Fi*/Isolated VLAN) for non-essential devices or IoT (Internet of Things) equipment limits the internal attack surface. Disabling unused ports and enabling *host-based firewalls* (such as UFW or Windows Defender Firewall) on end devices is also advised.

5.2. Wireless Security

Conclusion: The audit of the 802.11 spectrum, using tools such as airodump-ng and wifite, confirmed that the network is protected by WPA2/WPA3 encryption protocols. Although more secure than legacy protocols (WEP), they are susceptible to dictionary or brute-force attacks if the *handshake* is captured.

Recommendation: Ensure that the Wi-Fi network password has a high entropy (minimum of 16 characters, combining uppercase and lowercase letters, numbers and symbols) to prevent dictionary attacks (*offline cracking*). Additionally, features such as WPS (*Wi-Fi Protected Setup*) should remain disabled on *the ISP's* router.

5.3. Traffic Interception and Confidentiality

Conclusion: Packet capture performed through tcpdump and subsequent forensic analysis in NetworkMiner and Ettercap demonstrated how unencrypted traffic can be easily intercepted and read on a local network, exposing credentials, files and sessions.

Recommendation: It is imperative to adopt *Zero Trust* policies at the traffic level, ensuring that all sensitive communications use encrypted protocols (HTTPS, SSH, SFTP), discouraging or actively blocking the use of clean text protocols (HTTP, Telnet, FTP).

6. Command Log

Basic Recognition and Identification:

- Curl ipconfig.io
- IP A

Network Scanning and Discovery:

- sudo netdiscover -r [IP Redacted]
- sudo arp-scan [IP Redacted]
- sudo nmap -sn [IP Redacted]
- sudo nmap -Pn -T5 -f -p- [IP Redacted]
- sudo nmap -Pn -T5 -sU [IP Redacted]

Wireless Audit:

- iwconfig
- sudo airmon-ng start wlan0
- sudo airodump-ng --band abg wlan0
- sudo airodump-ng -c [Channel] --bssid [MAC_do_Router] wlan0 -w capture-wep
- sudo wifite -i wlan0 -wpa

Sniffing and Traffic Analysis:

- sudo arp -na
- sudo tcpdump -i eth0 -s 0 -U -w /tmp/captura.pca -C 500000
- ls -lha /tmp/captura.pca
- sudo ettercap -G
- sudo mono /opt/NetworkMiner_3-0/NetworkMiner.exe